Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/28/2008
PI Name:	Thompson Jr, James N. Ph.D.		
Project Title:	Genetic and Developmental Stability in Response to Long-Term Exposure of Drosophila melanogaster to a Space Station Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
PI Email:	jthompson@ou.edu	**Fax:**	FY 405-325-6202
PI Organization Type:	UNIVERSITY	**Phone:**	405-325-4821
Organization Name:	University of Oklahoma		
PI Address 1:	Department of Zoology		
PI Address 2:	RH 314		
PI Web Page:	http://faculty-staff.ou.edu/		
City:	Norman	**State:**	OK
Zip Code:	73019-3032	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	14	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Currently in Phase B for feasibility and strain characterization studies. NOTE: NCE to 9/30/2007, per A Chu 7/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Woodruff, Ronny (Bowling Green State University)		
Grant/Contract No.:	NCC2-1355		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using a simple invertebrate model, the fruit fly Drosophila melanogaster, we developed an experiment to study the effect of the space environment on: (1) successful reproduction over a period of seven generations on the International Space Station (ISS) in microgravity and 1 g control conditions; (2) genetic stability as measured by mutation accumulation during this multi-generation period; (3) the effect of microgravity, radiation, and other elements of the space environment on predicted reductions in developmental homeostasis (regulatory buffering) in response to environmental and/or mutational stresses; and (4) protein, DNA mutation, or RNA profiles from which stress mechanisms can be analyzed.. Mutation accumulation will use a genetic breeding program assay, Binscy, developed to allow successful accumulation of new mutations on the X chromosome with minimal personnel involvement. Mutation will be measured in genetic loci that cause an altered lifespan, that reduce viability (i.e., reduce relative fitness), that contribute to increased heritable variation in neurosensory organ number (principally sternopleural bristle number), and that yield DNA microsatellite and sequence variation. The experiment explored ways to assess changes in expression of genetic repair and other stress-related genes using proteomic profiles or RNA microarrays. The primary morphological characters assayed for developmental stability are changes in the pattern of wing support vein patterns and changes in the number and distribution of surface sensory organs (e.g., bristles and the campaniform sensillae located on the third longitudinal [L3] vein). A deviation from symmetry (fluctuating asymmetry, FA) measures the quality of developmental regulation in traits that are normally symmetrical. Together, these measures of genetic and developmental stability can extend our understanding of the rate and types of genetic changes that can occur during long-term exposure to a space environment and the effects of this novel stress on developmental homeostasis. This helps answer key questions about the ability of biological systems common to all organisms to adapt to novel environmental conditions of a space environment. Due to budgetary constraints, NASA decided to discontinue some of the basic gravitational biology research originally planned for the International Space Station, including development of the Drosophila Chamber hardware that would permit a multiple-generation Drosophila experiment on the ISS. But since the experiments we were doing to characterize strains and test the breeding programs and environmental stress effects have relevance beyond the ISS opportunity, we used the period since that notification to bring those experiments toward completion. We are pleased that phenomena associated with genetic effects on mutation and stress effects on developmental stability have been found to have broad application and to provide questions our lab will continue to follow in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of a long-term exposure to the radiation and gravitational condiditions of a space environment will extend our knowledge of the sensitivity and responsiveness of genetic and developmental systems to novel stresses. This information will help assess measures required for successful colonization and utilization of space.
	Using a simple invertebrate model, the fruit fly Drosophila melanogaster, we designed experiments to study the effect of the space environment on: (1) successful reproduction over a period of seven generations in the International Space Station (ISS) Insect Habitat in microgravity and 1 g control conditions; (2) genetic stability as measured by mutation accumulation during this multi-generation period; and (3) the effect of microgravity, radiation, and other elements of the space environment on predicted reductions in developmental homeostasis (regulatory buffering) in response to environmental and/or mutational stresses. Using a breeding program, Binscy, that was developed to minimize the need for personnel management during flight, mutation accumulation will be measured in genetic loci that cause an altered lifespan or that reduce viability (i.e., reduce relative fitness). We also measure mutational changes that cause nucleotide changes in selected DNA sequences or yield DNA microsatellite variation. The primary morphological characters assayed for developmental stability are changes in the pattern of wing support vein placement and changes in the number and distribution of surface sensory structures A deviation from symmetry (fluctuating asymmetry, FA) measures the stress response by means of reduced developmental coordination in traits that are normally symmetrical. Thus, in these sensory structures, significantly increased deviations from symmetrical bilateral expression indicate a reduction in the quality of developmental regulation or homeostasis. All of these experimental protocols were tested and verified during the grant period. The original plan was for experiments to be carried out in an insect habitat on the International Space Station over a period of approximately seven generations. But in April 2005 a change in experiment priorities led us to review and sharpen the experimental requirements for a multiple-generation experiment, and a series of options were developed as three alternative timelines that address a range of potential flight duration opportunities. Since a mutation accumulation assay requires multiple generations of exposure to the space environment conditions, such as microgravity, the highest priority was given to a 90-day experiment, although a 60-day or even 30-day experiment would yield valuable information. In addition we explored the use of using a stress-sensitive genotype and/or a somatic mutation assay to increase the sensitivity of the mutation rate assessments if flight duration makes the powerful Binscy mutation accumulation assay impractical. Characterization of these strains was done. We were notified by NASA Headquarters on 9 December 2005 and our grant administrators at Ames Research Center on 9 January 2006 that a change in NASA funding priorities would result in discontinuance of our planned multiple-generation mutation accumulation experiment on the International Space Station but with an opportunity to bring current experiments to completion. As a consequence of that work, we continued to characterize strains and protocols, mutation rate under stresses of vibration and hypergravity, assess stress response effects on developmental in genetically normal and in stress-sensitive and stress-resistant strains, and began pilot studies of proteomic changes resulting from vibration stress exposure, a technique tentatively planned for use on flies returned from the ISS. A key experiment planned for the International Space Station was the multi-generation mutation accumulation assay designed to allow new mutations to collect on a balanced (i.e., stably heterozygous) X chromosome, Binscy, named after the genetic markers it carries. This allows long-term mutation rate estimates to be made with minimum oversight by astronauts. On flies returned from the ISS, the assay allows an accurate measure of accumulated mutations of individually small effect on important life history traits like longevity and developmental stability and provides chromosomes in which DNA sequence, microsatellite, and other mutational events can be measured. For the on-ISS experiment, one source of data could come from measuring the changes predicted in relative proportion of genotypes from video images. As the habitat hardware and Drosophila Chambers were being developed at Ames Research Center, we assessed our ability to categorize sex and genotype using the camera system being tested there. Even though the camera system had not yet been optimized for the Drosophila habitat under development, the quality was sufficient if videoimages, not simply still images, are provided, since the flies moving through the focal plane essentially focus

Task Progress:	themselves. In these examples, the sex and eye phenotype were unambiguous, and images should only improve with refined video optimization and more experience. Mutation accumulation experiments with Binscy confirm it is an effective breeding program that yields results as predicted. Examples are provided graphically in the full Final Report. The use of mutation accumulation and control siblings raised together makes this a robust experimental design that overcomes many of the criticisms brought against other approaches to measuring genome mutation rate, U. It proves itself to be an excellent choice for an ISS experiment on genomic mutation. An extensive analysis was done of the repeatability and statistical analysis of using fluctuating asymmetry (FA) to measure stress effects on developmental coordination. Developmental stability has been defined as the group of processes that tend to resist or buffer the disruption of precise development. The resulting pattern is seen in random deviations from perfect bilateral symmetry. For a group of individuals, this is termed fluctuating asymmetry. With our traits, measurement error is very small compared to environmental stress effects; the system is robust and sensitive to the conditions being tested. In an intriguing experiment to assay developmental stability in offspring exposed to vibration and hypergravitational conditions, we found evidence of a beneficial effect of stress exposure. Vibration exposure is modeled after that experienced on the mid-deck of a shuttle during launch; hypergravity exposures were typically 5 ×g. Offspring from D. melanogaster females exposed to vibration stress fall into three categories by egg development phase at time of stress: eggs laid 1 to 3 days, 4 to 6 days, and 7 to 9 days after vibration of the female. Offspring from the first brood (nearly mature eggs) show variation in both positive and negative effects on offspring FA wing metrics as expected from random responses to physical factors. Offspring from the third brood have very few significant effects, as expected if stress responses have repaired any developmental effects generated from the vibration treatments. But the second brood is from eggs actively developing during the period most likely to reflect active stress response pathways. Strikingly, these offspring have wings displaying greater symmetry of shape following a brief exposure to vibration than the offspring from the unexposed control. The vibration stress appears to have caused a significant improvement in developmental stability of the offspring suggesting that a stress response is enhancing regulatory control of development. These are some of the potentially most important results of our whole NASA study, and we hope some future opportunity allows us to test these hypotheses in an extended space environment. They point toward a way of testing the relationship between stress exposure and a positive developmental or physiological response, a phenomenon called “hormesis”. Other positive benefits of stress pre-exposure have also been found in our experiments. Interestingly, in seven of the eight significant instances (88%) of a 5xg hypergravity times vibration interaction, the level of FA was reduced in the double treatment (hypergravity followed by vibration; it is not feasible to expose to both simultaneously in the lab). This suggests that exposure to one stress may stimulate a protective stress response that causes developmental stability to be improved when the same individuals are faced with a second stress. To provide representative DNA regions in which one can measure the frequency of nucleotide substitutions, deletions, or insertions under ISS space flight conditions, we selected four X-linked regions within well-characterized genes and one gene on the second chromosome implicated in proteomic assays as responding to stress conditions. Since the X-linked gene regions are present in a single copy in males, detection of nucleotide difference is uncomplicated by heterozygosity. Initial characterization of Canton-S, Binscy, and other potential ISS experimental strains for microsatellite genotypes was also carried out. Since microsatellite mutation rate is higher than that for typical genes, it can be a sensitive assay for some kinds of mutational event. But for the same reason, a strain will need to be re-characterized immediately before a flight experiment to confirm the strain’s homozygosity and current allelic makeup. For that reason, confirmation of the technique alone, not necessarily the genotype of all strains, was the primary goal at this time. That goal was met. In addition to providing foundation data to support potential short-term flight experiments if new opportunities arise and for possible experiments on evolution in space environment conditions, the experiments have pointed to interesting post-translational modifications of proteins coded by two genes, alcohol dehydrogenase (Adh) and yolk protein 1 (yp1), associated with vibration stress and stress-genome differences. Together with the identification of a protective effect of stress pretreatment and a beneficial effect of stress exposure in activating hormesis-like stress responses that improved developmental stability (reduced developmental asymmetry), these lines of research will continue for the foreseeable future in our labs. Results will support plans for future tests of these hypotheses if flight opportunities again arise.
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